Roll No. Total No. of Pages :02

Total No. of Questions : 13

B.Pharmacy (Sem.-5)

PHARMACOGNOSY AND PHYTOCHEMISTRY II-THEORY

Subject Code : BP504T

M.Code :76789

Date of Examination: 01-12-2025

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carryingTEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Write briefly:**

   a) What are primary and secondary metabolites?

   b) Which biosynthetic pathways are responsible for the biosynthesis of Cardiac glycosides?

   c) Give the biological source, active constituents and uses of a plant containing Anthraquinone glycosides.

   d) Give the biological source and uses of 2 plants containing volatile oils.

   e) Compare the structure of Atropine and Hyoscine.

   f) What is the advantage and limitation of Ultrasound assisted extraction?

   g) What information do we get from the IR spectra of a phytoconstituent?

   h) What are Resins? Give an example of a plant which contains resins.

   i) If a plant extract gives negative test with Mayer's reagent and a positive Murexide reaction - give the structure of the phytoconstituent present in it.

   j) How are saponins classified?

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SECTION - B

2. Describe in detail the biosynthesis of Flavonoid glycosides.

3. What are alkaloids? How are alkaloids classified? What is the medicinal importance of Opium alkaloids?

4. What is the importance and applications of chromatographic techniques in Pharmacognosy?

SECTION - C

5. How are carotenoids biosynthesised?

6. What are the different methods used in the investigation of biosynthetic pathways?

7. Write a detailed note on medicinal and commercial uses of Digitalis.

8. What are the advantages and challenges of using Artemisia or Taxol?

9. What is electrophoresis? What are its applications?

10. Describe the industrial production and utilization of Diosgenin.

11. Write a note on the importance of 'Green methods' of extraction.

12. What are tannins? How are they classified? What are the medicinal and commercial uses oftannins?

13. Describe the methods used in isolation and identification of Rutin or glycyrrhetinic acid. Give the biological source and uses also.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

2025